

## Advanced (Habanero)

**Q1.** Convert  $x^2 + 6x + 9$  to vertex form.

- (A)  $(x + 3)^2$
- (B)  $(x - 3)^2$
- (C)  $(x + 3)^2 + 3$
- (D)  $(x - 3)^2 - 3$
- (E)  $(x + 3)^2 - 3$

**Q2.** Complete the square for  $x^2 + 4x$ .

- (A)  $(x + 2)^2$
- (B)  $(x - 2)^2$
- (C)  $(x + 2)^2 - 4$
- (D)  $(x - 2)^2 + 4$
- (E)  $(x + 2)^2 + 4$

**Q3.** What is the vertex of the parabola  $y = (x - 2)^2 + 1$ ?

- (A) (0, 1)
- (B) (2, 0)
- (C) (2, 1)
- (D) (-2, 1)
- (E) (1, 2)

**Q4.** Convert  $x^2 - 8x + 16$  to standard form.

- (A)  $(x - 4)^2$
- (B)  $(x + 4)^2$
- (C)  $(x - 4)^2 - 16$
- (D)  $(x + 4)^2 + 16$
- (E)  $(x - 4)^2 + 16$

**Q5.** Complete the square for  $x^2 - 2x$ .

- (A)  $(x - 1)^2$
- (B)  $(x + 1)^2$
- (C)  $(x - 1)^2 - 1$
- (D)  $(x + 1)^2 + 1$
- (E)  $(x - 1)^2 + 1$

**Q6.** What is the axis of symmetry of the parabola  $y = (x + 1)^2 - 2$ ?

- (A)  $x = -1$
- (B)  $x = 1$
- (C)  $x = 0$
- (D)  $x = 2$
- (E)  $x = -2$

**Q7.** Convert  $x^2 + 2x + 1$  to vertex form.

- (A)  $(x + 1)^2$
- (B)  $(x - 1)^2$
- (C)  $(x + 1)^2 + 1$
- (D)  $(x - 1)^2 - 1$
- (E)  $(x + 1)^2 - 1$

**Q8.** What is the minimum value of the function  $y = (x - 3)^2 + 2$ ?

- (A) -2
- (B) 0
- (C) 2
- (D) 3
- (E) 5

**Open Ended Questions (FRQ)****Q9.** Convert the equation  $x^2 - 6x + 8$  to vertex form and find the vertex of the parabola.**Q10.** Complete the square for the equation  $x^2 + 4x - 3$  and write the equation in standard form.**Chef's Tips**

- Tip 1: Ensure students understand the concept of completing the square.
- Tip 2: Remind students to check their work by plugging in values.
- Tip 3: Encourage students to use graphing calculators to visualize the parabolas.